Experiment No: 3 – Expanding and Squeezing NumPy Arrays

Aim

To study expanding, squeezing, and sorting operations on NumPy arrays.

Software Required

Python 3.x
NumPy Library
Jupyter Notebook / Google Colab / Python IDLE

Theory

NumPy provides functions to expand dimensions, remove singleton dimensions, and sort arrays.

a) Expanding a NumPy Array

Description.

Syntax

np.expand_dims(array, axis)

Program

```
import numpy as np
arr=np.array([10,20,30])
print(np.expand_dims(arr,0))
```

Output

[[10 20 30]]

b) Squeezing a NumPy Array

Description.

Syntax

np.squeeze(array)

Program

```
import numpy as np
arr=np.array([[[1,2,3]]])
print(np.squeeze(arr))
```

Output

[1 2 3]

c) Sorting in NumPy Arrays

Description.

Syntax

np.sort(array)

Program

```
import numpy as np
arr=np.array([5,2,9,1])
print(np.sort(arr))
```

Output

[1 2 5 9]

Result

Thus, the required NumPy operations were successfully performed.